REVIEW

Formulation Considerations for Proteins Susceptible to Asparagine Deamidation and Aspartate Isomerization

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ABSTRACT: The asparagine (Asn) deamidation and aspartate (Asp) isomerization reactions are nonenzymatic intra-molecular reactions occurring in peptides and proteins that are a source of major stability concern in the formulation of these biomolecules. The mechanisms for the deamidation and isomerization reactions are similar since they both proceed through an intra-molecular cyclic imide (Asu) intermediate. The formation of the Asu intermediate, which involves the attack by nitrogen of the peptide backbone on the carbonyl carbon of the Asn or the Asp side chain, is the rate-limiting step in both the deamidation and the isomerization reactions at physiological pH. In this article, the influence of factors such as formulation conditions, protein primary sequence, and protein structure on the reactivity of Asn and Asp residues in proteins are reviewed. The importance of formulation conditions such as pH and solvent dielectric in influencing deamidation and isomerization reaction rates is addressed. Formulation strategies that could improve the stability of proteins to deamidation and isomerization reactions are described. The review is intended to provide information to formulation scientists, based on protein sequence and structure, to predict potential degradative sites on a protein molecule and to enable formulation scientists to set appropriate formulation conditions to minimize reactivity of Asn and Asp residues in protein therapeutics. J Pharm Sci 95:2321–2336, 2006

Keywords: chemical stability; proteins; peptides; formulation; deamidation

INTRODUCTION

With the advent of recombinant DNA technology, protein drugs have evolved into a major class of human therapeutics. The emergence of this technology has enabled large-scale manufacturing of safe and efficacious protein products. This technology, when coupled to protein engineering techniques, provides additional advantages in the development of "engineered" proteins. Engineered proteins are modifications of the native (wild-type) protein forms in which specific amino acid residues are altered for the purpose of improving physicochemical stability and activity, and reducing immunogenicity of the native proteins.[1] Examples of currently used protein-based drug therapies include the treatment of malignant lymphomas, colorectal carcinomas, allergic asthma, ulcerative colitis, heart disease, and neurological conditions.[1–6]

Recent advances in the production of protein therapeutics have necessitated improvements in the area of protein formulation development to maintain the safety and stability and, thereby, the efficacy of a protein molecule from the time of its production to its use. Under physiological

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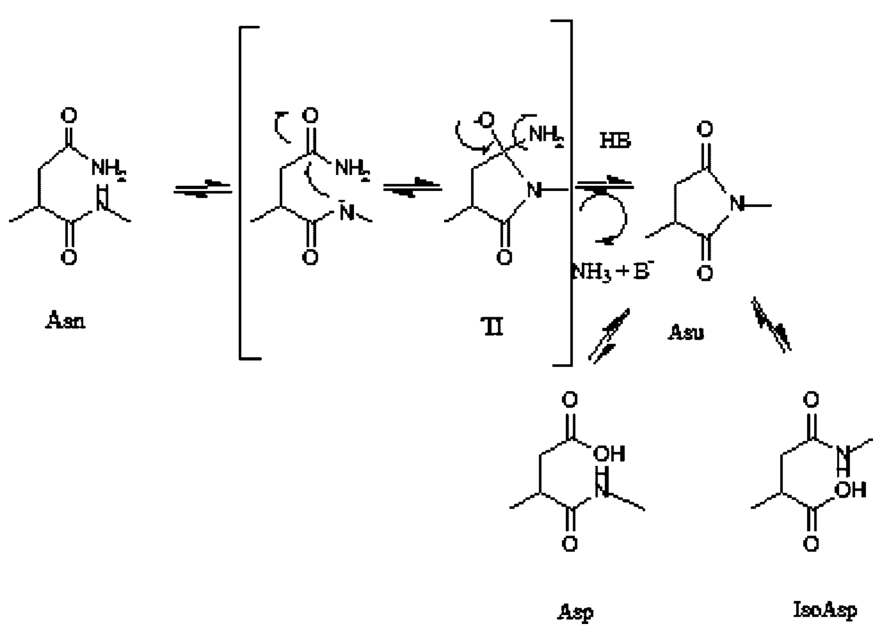
conditions, proteins spontaneously degrade via physical and chemical degradative processes.[7] The manufacturing and storage conditions for a protein product generate physical and chemical stresses that can compromise the stability of the protein in a therapeutic product.[8] The non-enzymatic post-translational modification of both Asn and Asp residues (Asx) is one of the major chemical degradative pathways for proteins during production and storage.[9–14] The Asx residues in proteins can degrade through formation of an intra-molecular succinimide (Asu) that leads to the formation of protein variants. These variants are proteins bearing the Asp, IsoAsp, or Asu residues at the site of the labile Asn residues and the IsoAsp or Asu residues at the site of the labile Asp residues.[11,12,15–20] The variants resulting from Asn or Asp degradation can have a significant impact on the safety and the efficacy of a pharmaceutical protein product.[21–24] The Asn and Asp degradation can also lead to molecular heterogeneity, which can present challenges to the manufacturing of a consistent protein product. Therefore, thorough identification and characterization of the proteins susceptible to Asn deamidation and Asp isomerization reactions and the protein variants resulting from these degradative reactions is essential for development of stable, efficacious, and consistent protein therapeutics.

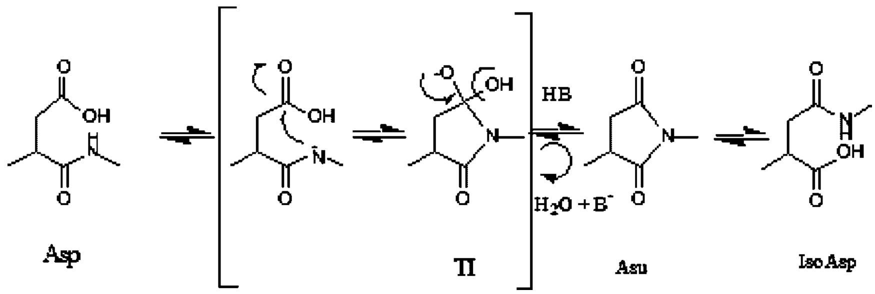
DEGRADATIVE PATHWAY OF Asn DEAMIDATION AND Asp ISOMERIZATION IN PEPTIDES AND PROTEINS

The degradative pathways for Asn deamidation and Asp isomerization reactions at pHs >5 are as shown in Schemes 1 and 2, respectively.

Both the deamidation and isomerization reactions proceed via ionization of the peptide-bond nitrogen located on the C-terminal side of the Asx residue. This is followed by nucleophilic attack of this ionized peptide-bond nitrogen on the carbonyl carbon of the Asx side chain to form the oxyanionic tetrahedral intermediate (TI), as shown in Schemes 1 and 2. In the case of Asn deamidation, protonation of the amine group of the TI and the subsequent loss of ammonia lead to the Asu intermediate (Scheme 1). The Asu intermediate, as shown in Scheme 2 can also form from the TI during Asp isomerization. The TI in the Asp isomerization pathway bears a hydroxyl leaving group that, upon protonation and subsequent loss of a water molecule, leads to the Asu intermediate. The formation of the Asu intermediate constitutes the rate-limiting step for both the deamidation and the isomerization pathways. The Asx racemization reaction is believed to occur at the stage of the cyclic imide.[16] However, Li et al.[15] have shown that racemization can also occur at the stage of the TI in case of the Asn deamidation. The



Scheme 1. Degradative pathway for Asn deamidation in peptides and proteins at pH > 5.



Scheme 2. Degradative pathway for Asp isomerization in peptides and proteins at pH > 5.

cyclic imide then undergoes rapid hydrolytic cleavage at the α or β carbon to form the isoaspartate (iso-Asp)- or the aspartate (Asp)-containing products, respectively, in a ratio of approximately 3 or 4:1. Peptide bond hydrolysis between Asx and its succeeding residue can occur along with the deamidation and isomerization reactions at pH ≤ 5. The reaction rates for the peptide bond hydrolysis are significantly inhibited at pH > 5.[12]

The mechanism of Asn deamidation and Asp isomerization has been extensively studied in the solid state.[25–27] The pathway for Asp isomerization has been observed to be similar in both solution and solid states.[27] The formation of the Asu intermediate was the rate-determining step for the isomerization reaction in both solution and the solid state. Studies of Asn deamidation in the solid state have shown that differences exist between the Asn deamidation mechanism in solution and the solid state at formulation pH > 8. The formation of the Asu intermediate during Asn deamidation comprises two sub-steps: (1) the formation of the TI and (2) the loss of a molecule of ammonia from the TI to form the Asu intermediate. In studies of reaction mechanisms in the solid state, it has been shown that Step 2 is the rate-limiting step for the deamidation reaction, whereas Step 1 is rate limiting for the reaction in solution.[25]

pH DEPENDENCY OF Asn DEAMIDATION AND Asp ISOMERIZATION REACTIONS

Patel et al.[17] and Oliyai et al.[19] have extensively studied the pH dependency of deamidation and isomerization reactions, respectively, in the pH range of 1 to 12. In this section, the pH dependency of deamidation and isomerization reactions is discussed within the pH range of 5–8 since proteins are typically formulated within this narrow pH range. Formulating proteins in more acidic or basic solutions can lead to protein unfolding.[28] Because proteins are typically administered either intravenously (IV) or subcutaneously (SC), formulation pH > 8 is not ideally suited for a commercial product. Formulations at pH > 8 also render the proteins more susceptible to base-catalyzed degradations such as deamidation.[17] The pH-rate profiles for Asn deamidation and Asp isomerization reactions in solution, in the pH range of 5 to 8, are shown in Figure 1. The data for generating these profiles were obtained from investigations conducted with model hexapeptides VYPNGA[17] and VYPDGA.[19]

Based on the pH-rate profile for the deamidation reaction, as shown in Figure 1, it was observed that the deamidation reaction is base-catalyzed within the pH range of 5 to 8. The deamidation rate was proportional to the hydroxide ion concentration of the solution. Increasing hydroxide ion concentrations in solution lead to faster reactivity of the Asn residue. The reaction mechanism (Scheme 1) indicates that increasing hydroxide ion concentration would lead to increased ionization of peptide backbone, resulting in increased Asn deamidation rates.

The potential catalytic role of acetate, phosphate, and Tris buffers on Asn deamidation reaction was investigated in VYPNGA.[17] In these investigations, buffer catalysis due to acetate (pH 5) and phosphate (pH 6.0 and 6.5) buffers was not observed. However, buffer catalysis due to phosphate and Tris buffers at formulation pH ≥ 7 was observed. Based on these studies, it can be concluded that the deamidation reaction, in solution, is specific base-catalyzed within the pH range 5 < pH < 7 and general base-catalyzed at pH > 7.[17]

The pH-rate profile for Asp isomerization shown in Figure 1 was obtained from studies conducted

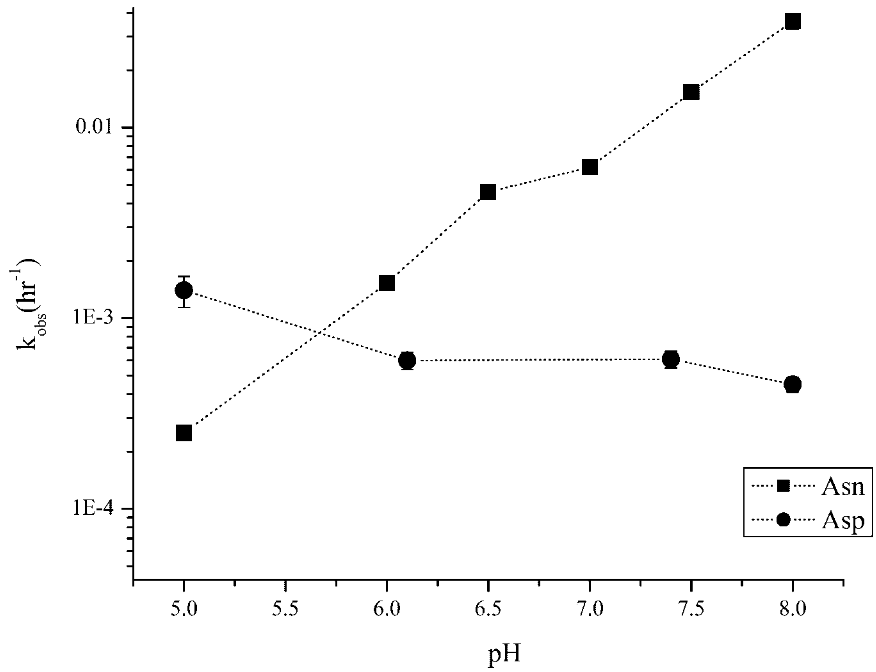


Figure 1. The pH dependency of Asn deamidation (■) and Asp isomerization (●) reactions in aqueous solutions maintained at 37°C and $I = 0.5$ M. Data for this plot were obtained from kinetics studies conducted with model peptides VYPNGA and VYPDGA.

with the peptide model VYPDGA.[19] In the pH range of 5–6, the reaction is acid catalyzed. The Asp isomerization rate decreases with increasing hydroxyl ion concentration in the pH region of $5 < \text{pH} < 6$. In this pH region, that is, at pH > pKa of the Asp side chain (pKa ~ 4), the increasing hydroxyl ion concentration leads to an increased ionization of the Asp side chain. The ionized Asp side chain, the carboxylate form, provides for a weaker leaving group than the protonated side chain, the carboxylic form, in the isomerization pathway (Scheme 2). The ionized side chain inhibits the rate of formation of the cyclic imide, which is the rate-limiting step, leading to a decrease in the overall isomerization rate. At pH > 6, the increase in the hydroxyl ion concentration aids in the increased ionization of the peptide backbone nitrogen. This increased ionization of the peptide backbone would accelerate the formation of the Asu intermediate, thereby increasing isomerization rate. These two opposing factors, one being the increased ionization of the peptide backbone and the other being the increased ionization of the Asp side chain, offset each other, and the overall isomerization rate is thus observed to be pH independent at pH > 6. The potential role of buffers in catalyzing Asp isomerization was evaluated in the pH range of 5 to 8. These investigations indicated the lack of a catalytic role of acetate, phosphate, or Tris buffers on Asp isomerization in the hexapeptide.[19] The lack of general base catalysis at pH > 7 for the Asp isomerization reaction is in contrast to the general base catalysis observed for the Asn deamidation reaction. This lack of general base catalysis for the isomerization pathway indicates a distinct change in the kinetic behavior of the reactions upon replacing an Asn with an Asp in a hexapeptide.

The dependency of Asn deamidation and Asp isomerization reaction rates on the effective pH ("pH") in the solid state was evaluated.[25,27] The term "pH" of a lyophilized formulation was defined as the pH of the solution from which it is lyophilized. Studies of deamidation rates of VYPNGA in glassy and rubbery polymeric solids formulated using poly(vinyl pyrrolidone) (PVP) in the "pH" range 5 to 8 demonstrate an attenuation in the deamidation rates of the peptide in a lyophilized solid due to possible reduction in molecular mobility in these lyophilized formulations.[25] The deamidation rates of VYPNGA in PVP solutions were ~10 times faster than the deamidation rates of the peptide in rubbery solids formulated using PVP. Also, the deamidation rates of VYPNGA in rubbery solids containing PVP were ~10 times faster than the deamidation rates in glassy solids containing PVP. The deamidation rate in the solid state was observed to be base-catalyzed in the "pH" range 5 to 8, which was in agreement with the mechanism of Asn deamidation in solution. However, the slope of the "pH" rate profile for Asn deamidation in the solid state for the "pH" range 5 to 8 is less than the slope for this profile in solution. This suggests the deamidation rates to be less dependent on hydroxide ion concentration in the solid state. A change in the physical state of the formulation from solution to a lyophilized solid also influenced the pH dependency of Asp isomerization rates in the peptide VYPDGA.[27] In lyophilized formulations of lactose/peptide and mannitol/peptide, the peptide VYPDGA showed a decrease in the pH dependency of the Asp isomerization rates.[27] A decrease in the overall rates of Asp isomerization observed in the solid state was attributed to a decrease in the conformational flexibility of the peptide backbone and the Asp side chain in a lyophilized formulation.[27] The studies with peptide models VYPNGA and VYPDGA in the solid state demonstrate that the change in physical state of the formulation does not alter the mechanism and the pH dependency of deamidation and isomerization reactions within the pH range of 5 to 8.

In recent investigations in our laboratory, the influence of a change in physical state of the formulation on the catalytic effect of an adjacent His residue on Asn deamidation rates in GQNHH was evaluated (unpublished work). This research has demonstrated that the acid catalysis due to the C-terminal His residue, observed for GQNHH deamidation in solution,[29] was substantially inhibited upon formulating the peptide in lyophilized matrix. This result suggests that in peptides and proteins wherein catalytic residues lie in close vicinity of a labile Asn residue, formulating these biomolecules in a lyophilized form may provide additional advantages in achieving peptide/protein stability.

EFFECT OF FORMULATION VARIABLES ON Asx REACTIVITY

Significant differences in the dependency of Asn deamidation and Asp isomerization reactions on extrinsic factors such as pH and solvent/formulation dielectric has been reported.[12,17,19,30] The differences in the dependency of these reactions on extrinsic factors are a result of differences in the nature of the labile Asn and Asp residues. The Asp residue, which is an ionizable residue, exists in the carboxylic acid and the carboxylate forms; these forms have significantly different susceptibilities to the isomerization pathway.[30] As discussed earlier, the carboxylate form of the Asp is stable to the isomerization reaction whereas the carboxylic acid form is the reactive species.[19] Therefore, formulation conditions that would favor the carboxylate form of the Asp, as opposed to the carboxylic acid form, would render the residue less susceptible to isomerization reaction. The effects of various formulation conditions on both Asn and Asp reactivity have been summarized in Table 1 and will be discussed in detail in the following sections.

pH of the Formulation

As discussed in an earlier section, the pH of the protein solution has a significant impact on the reactivity of the Asx residues in the protein. At $pH > 5$, an increase in the formulation pH leads to an increase in Asn deamidation rates. The pH range 3 to 5 offers the region of optimum stability to Asn deamidation for a protein or peptide species.[17,31] However, formulating a protein under these pH conditions is not suitable due to concerns about protein aggregation and hydrolysis.[28] As a consequence, lower pH ($5 < pH < 6$) are well suited for formulation of protein products for which Asn deamidation is anticipated to be a major degradative pathway. In the case of Asp isomerization, formulation at $pH > 6$ leads to a decrease in Asp reactivity. In most proteins where Asp isomerization is a critical stability issue, the desired pH range for formulation would be $6 < pH < 8$.[19]

The "pH" dependency of Asx reactivity has also been extensively studied in lyophilized solids.[25–27] The term "pH" is widely used for solid state formulations because defining the pH in a solid state is difficult due to complications in measuring hydrogen ion activity in the solid state. Several methods have been employed to directly measure hydrogen ion activity in the solid state including nuclear magnetic resonance spectroscopy (NMR),[32,33] fourier transform infrared spectroscopy (FTIR),[34] and electron paramagnetic resonance spectroscopy (EPR).[35] The results from these investigations demonstrate that lyophilized formulations exhibit a pH memory, that is, the pH in the solid formulations is identical to the pH of

Table 1. Formulation Considerations for Proteins Undergoing Asn Deamidation and/or Asp Isomerization

Formulation Variable	Asn Deamidation	Asp Isomerization
pH	The deamidation reaction is a base-catalyzed reaction in the pH range 5 to 8. Optimal pH range for stability of a protein to deamidation is between pH 3 and 5.[17]	The isomerization reaction is acid catalyzed at $4 < pH < 6$. Optimal pHs for stability of a protein to isomerization are >7.[19]
	For formulation purposes, pHs close to 5 are ideal	For formulation purposes, ideal pHs are $6 < pH < 8$
	General base catalysis of deamidation rates was observed with phosphate, Tris, and carbonate buffers at pHs >7.[17]	No evidence for a significant degree of buffer catalysis has been reported.
Temperature	The deamidation rates followed an Arrhenius relationship with temperature. The activation energy of 21.7 kcal/mole has been reported for the deamidation reaction in literature[16,17]	An Arrhenius relationship between the isomerization rates and storage temperature was also observed for Asp isomerization with an activation energy of 21.7 kcal/mole.[16]
Solvent dielectric	A decrease in solvent dielectric from 74.2 (100% water, 37°C) to 46.1 (80% w/w glycerol in water, 37°C) led to a ~ sixfold decrease in reactivity[40]	A decrease in the solvent dielectric from 69.8 (100% w/w water, 50°C) to 44.7 (80% v/v glycerol in water, 50°C) led to a ~ twofold increase in reactivity for a peptide and a ~ fourfold increase in reactivity for a protein (unpublished work).
	Formulations incorporating organic co-solvents in solution formulations may help increase stability to deamidation.	Formulations incorporating organic co-solvents in solution formulations may increase reactivity of Asp residues.
Solvent viscosity	Increases in formulation viscosity from 0.6 cp to 29000 cp achieved using PVP-based aqueous solutions and hydrated PVP solids led to ~ 10-fold reduction in deamidation rates.[42]	No studies have been reported for effects on solvent viscosity on Asp isomerization.
	The viscosity effect on deamidation rates was more significant at higher PVP concentrations (>50% w/w) whereas at lower PVP concentrations (0–20% w/w), the effect of solvent dielectric of the PVP solutions on deamidation rates was predominant[41,42]	Theoretically, the predicted effect of solvent viscosity on Asp isomerization rates would be similar to what has been reported for Asn deamidation rates, with increasing viscosities leading to lowering of Asp reactivity.
Physical state	Formulating in lyophilized PVP matrices in glassy state led to a 10000-fold decrease in deamidation rates compared to the rates in PVP solutions[25]	Formulating in lyophilized lactose led to an increased stability to the isomerization reaction under different moisture and temperature conditions studied.[27]
	Increased moisture content in these PVP solids led to increased deamidation rates.[25]	

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the aqueous solutions from which they were lyophilized.[34]

The "pH"/pH dependency of reaction rates is similar in the solid state and the solution state for Asn deamidation[25,36] and Asp isomerization[27] reactions. This suggests that pH conditions that stabilize peptides and proteins to Asx reactivity in solution can also be employed to maintain Asx stability in the solid state.

Temperature of Formulation Storage

The stability of the formulated proteins to deamidation and isomerization reactions is dependent on the temperature of storage of the formulations. An increase in temperature leads to an increase in reactivity of Asx residues in proteins. The Asx reaction rates of peptides exhibited an Arrhenius relationship with temperature in solution.[16,17] The Arrhenius nature of the Asx reactivity–temperature relationship suggests that accelerated stability studies investigating peptide/protein deamidation or isomerization at elevated temperatures can be used to predict protein deamidation or isomerization rates under storage conditions for a peptide/protein drug product. In the solid state, the deamidation rates were observed to be less temperature dependent than those in solution.[26]

Storage temperature can indirectly influence Asx reactivity by affecting the ionic equilibria of the buffer components, which results in shifts in the pH of the formulations.[37] Amine buffers (Tris, histidine, etc.) have high temperature coefficients[37] and, therefore, storage at temperatures different from the temperature of formulation preparation could shift the pH of the formulations. Since deamidation and isomerization reactions are pH-sensitive processes, these shifts in formulation pH could potentially enhance the reactivity of the Asx residues. Another indirect result of temperature is its effect on hydroxyl ion concentration important in base-catalyzed reactions such as deamidation, which are dependent on hydroxyl ion concentration. The dissociation constant of water, which is a function of hydrogen ion and hydroxyl ion concentration, varies with temperature. As a result of the variation of the dissociation constant, the hydroxyl ion concentration of water can vary as a function of temperature.[37]

The above observations concerning the influence of temperature on Asx reaction rates suggest that long-term storage of protein formulations at lower temperatures would be ideal for maintaining protein stability to deamidation and isomerization reactions. Also, temperature control of formulations during shipping and storage is more important for protein solutions than for lyophilized proteins. In case of lyophilized formulations, storage of the formulations at temperatures below the Tg of the lyophilized cake is recommended.

Excipient Effects

Excipients are commonly added to protein formulations for the purpose of improving protein stability and as formulation bulking agents. Some examples of excipients added to protein formulations include sucrose, trehalose, mannitol, glycerol, Tweens, buffer salts (histidine salts, phosphate salts), and ionic strength modifiers (sodium chloride). The addition of these excipients to protein formulations can impact reactivity of the Asx residues through affecting the pH, dielectric constant, and viscosity of the formulation. Also, the addition of excipients can potentially affect the conformation of a protein, thereby influencing reactivity of Asx residues in a protein.

Buffers are excipients added to a protein formulation for the purpose of pH control in a formulation. However, the inclusion of buffers at higher concentrations in a protein formulation can adversely affect the stability of Asx residues in a protein as was discussed in the section titled "pH Dependency of Asn Deamidation and Asp Isomerization Reactions," where general base catalysis of Asn deamidation rates was observed with Tris and phosphate buffers at pH ≥ 7.0.[17,38] Amine buffers have high temperature coefficients which renders the formulation using amine buffers susceptible to variations in pH due to changes in temperature. The variation in pH resulting from the effect of temperature on amine buffer pKa's can influence the reactivity of the Asx residues in proteins. Thus, careful consideration of catalytic contributions and the temperature coefficients of buffer components is necessary for the choice of suitable buffer systems in a protein formulation.

Excipients such as glycerol, alcohols, and sucrose are routinely added to protein formulations for the purpose of improving protein physical stability.[39] The use of these excipients can significantly impact the solvent dielectric of the formulation. The effects of solvent dielectric of alcohol- and glycerol-based aqueous formulations on degradation rates of Asx residues in model peptides has been studied by Brennan et al.[30] These studies observed that the rate of Asn deamidation was significantly inhibited in alcohol/water and glycerol/water solvent systems due to a reduction in the dielectric strength of water caused by addition of co-solvents such as alcohol and glycerol. The decreased reactivity of Asn residues in peptides in solvents of lower dielectric strength was attributed to decreased stabilization of the ionic intermediates formed during the cyclization step of the Asn deamidation pathway.

It is interesting to note that a similar effect of solvent dielectric on reaction rates of Asp residues was not observed. A closer examination of the reaction pathway for Asp isomerization reveals that, in addition to the equilibrium between the Asp and the IsoAsp forms of the peptide, there also exists an equilibrium between the carboxylic acid and the carboxylate form of the Asp side chain. The latter equilibrium is dependent on the pKa of the Asp. It was presumed that increased concentrations of co-solvent led to an increase in the pKa of Asp and thereby increased concentrations of the carboxylic acid form of the Asp side chain at physiological pH. This carboxylic form of the Asp side chain is more reactive to the isomerization reaction. The authors presumed that the retardation in cyclization rates, as noted for the deamidation pathway, was offset by the increased reactivity of the Asp side chain. This resulted in a reduced effect of solvent dielectric strength on Asp isomerization rates.[30] However, studies on a recombinant humanized monoclonal antibody (MAb I) and its peptide model VDYDG have demonstrated that the effects of solvent dielectric on Asp reactivity, although less pronounced than those reported for Asn reactivity,[40] are significant (unpublished work). These studies show that a decrease in the solvent dielectric due to addition of co-solutes such as glycerol, sucrose, and ethanol to a peptide/protein solution leads to a significant acceleration in the rates of Asp isomerization of these biomolecules.

The effect of solution viscosity and solution polarity on the deamidation rates of VYPNGA was explored by varying the concentrations of glycerol and PVP in aqueous formulations of the peptide.[41] A multidimensional free energy model was used to study the effect of solution viscosity and solvent dielectric on deamidation rates of Asn residue in the peptide. Regression analysis performed on the data using the free energy model showed that the effect of solvent dielectric on the deamidation rates of the peptide was observed to be most significant. A decrease in the dielectric strength of the medium from 80 (water) to 35 (PVP/glycerol/water formulations) led to an approximately sixfold decrease in peptide deamidation rates.[41] The effect of solution viscosity on Asn deamidation rates in model hexapeptides was evaluated within the viscosity range of 0.6 to 29000 cp. This wide range in solution viscosity was achieved through varying the concentrations of PVP in aqueous solutions; the PVP solutions $\sim$29000 cp viscosity were obtained by hydrating the lyophilized PVP solids.[42] These studies suggest that effects of macroscopic viscosity on Asn deamidation rates are more significant at viscosities $\sim$29000 cp. However, one limitation of this study was that the effect of concentrated PVP solutions on the solvent dielectric of water was not addressed. The increase in PVP concentrations of solutions would lead to decrease in the solvent dielectric.[41] The tenfold decrease in deamidation rates observed at high PVP concentrations ($\sim$80% w/w) in solution may not be entirely a viscosity effect. The effect of the concentrated PVP solutions in decreasing the solvent dielectric and thereby influencing Asn reactivity needs to be considered.

Stabilizers such as sucrose and glycerol are routinely added to protein formulations to improve stability of proteins during lyophilization or storage.[28,39,43,44] A decrease in reactivity of cysteine residues located on the protein surface was observed in sucrose-containing protein formulations.[43] This was attributed to a reduction in the conformational mobility of the surface residues in the protein upon addition of sucrose to a protein formulation, decreasing reactivity of cysteine residues.[43] Retardation of Asn deamidation rates in the model hexapeptide VYPNGA upon addition of sucrose to the solution formulation of VYPNGA was observed.[45] The observed decrease in Asn deamidation rates in the presence of sucrose in the formulation was attributed to a greater conformational preference of the peptide for a type II β-turn structure in which the peptide was less prone to Asn deamidation.[45] However, the addition of sucrose could also result in a decrease in the dielectric constant of the formulation. This decrease in solution dielectric constant could also lead to an inhibition of deamidation rates. This effect of sucrose on the solution dielectric constant and thereby the deamidation rates of the peptides was not addressed in this study.

It is well known that the conformational flexibility of Asx residues is critical for cyclization of these residues during the deamidation and isomerization reactions.[12] Excipients that affect the conformational flexibility of these residues in a protein could therefore have significant influence on their reactivity.

Metal ions can selectively bind to different sites on a protein molecule which in turn can have a significant impact on protein stability and activity. The effect of metal ions on protein structural stability and activity has been studied in proteins such as calmodulin and insulin.[46–48] These investigations have reported increased structural stability and activity of these proteins in formulations containing zinc and magnesium ions. Addition of magnesium salts to humanized monoclonal antibody formulations has produced an improvement in the stability of the antibody to Asp isomerization.[49] Also, in recent investigations in our laboratory, the effect of addition of sodium chloride, magnesium chloride, and calcium chloride salts to formulations on Asp reactivity of MAb I and VDYDG was evaluated (unpublished work). These results have shown that addition of salts to aqueous formulations of these biomolecules has no influence on Asp reactivity. However, in glycerol-based formulations of MAb I and VDYDG, a significant inhibition in the rates of Asp isomerization upon salt addition was observed for both biomolecules.

Physical State of the Formulation (Lyophilized Solids)

Formulating a protein in the solid state is advantageous as it prolongs the shelf-life of a protein therapeutic.[39] Although the reactivity of Asx residues is greatly reduced in proteins formulated as lyophilized products in some lyophilized formulations, the reactivity of Asx residues could still be significant enough to compromise protein stability.[50] A detailed discussion on solid-state chemical stability of peptides and proteins is available in articles presented by Lai et al.[50] and Byrn et al.[51] Asx reactivity has been extensively studied in detail in both aqueous[11,12,16,20,41,42,52] and lyophilized formulations.[25,36,50,53,54] Investigations of Asx reaction rates in both solution and solid state in various model hexapeptides containing labile Asx residues have demonstrated that a change in the physical state of a formulation from a solution to a lyophilized solid significantly influences Asx reactivity. The reactivity of Asx residues in a protein or a peptide formulated as a lyophilized solid is affected by three major factors: (1) a reduction in the mobility of both the peptide backbone, and the side chains,[26,27,55] (2) a reduced water content,[54] and (3) an alteration in the pH dependency of the reaction[25,56] in the solid state at pH > 8. In studies conducted with a model hexapeptide VYPNGA, a 10000-fold decrease in Asn deamidation rates was observed in glassy polymeric solids prepared using PVP as the bulking agent compared to the deamidation rates of the peptide in PVP-based solutions (5% w/v).[25] Studies conducted with VYPNGA in mannitol- and sucrose-based formulations show 2- and 80-fold reductions, respectively, in Asn deamidation rates of the peptide in lyophilized solids when compared to deamidation rates of this peptide in solutions containing these carbohydrates.[55] Studies with the model hexapeptide VYPNGA in PVP solutions and lyophilized solids obtained from these PVP solutions suggest that a change in reaction mechanism occurs as the physical state of the formulation is changed from a solution to a lyophilized solid. The deamidation reaction mechanism changed from that of a general base-catalysis mechanism to that of a pH-independent process at pH > 8, as the physical state was changed from solution to lyophilized solid.[25] This change in the rate-determining step of the Asn deamidation reaction that occurs at pH > 8 in the solid-state studies with the peptide VYPNGA was also noted in studies exploring Asn deamidation in other pentapeptides.[56] Also, studies with VYPDGA have reported that inclusion of lactose, an amorphous bulking agent, in peptide formulations imparted a stabilizing effect to the peptide at different moisture levels and temperature conditions studied.[27] The inhibition in deamidation and isomerization rates of peptides is primarily due to mobility constraints encountered by the peptide or protein in a solid matrix. The addition of excipients such as PVP that increase the glass transition temperature (Tg) of the lyophilized solids would greatly enhance the stability of Asx residues to deamidation and isomerization reactions in the solid state.[54,55]

EFFECT OF PRIMARY SEQUENCE ON Asx REACTIVITY

The effects of the primary sequence of a protein on Asx reactivity have been extensively investigated using various model pentapeptides and hexapeptides.[7,11,57–60] In solution, the amino acid residue in the position C-terminal to a reactive Asx residue has been reported to have a greater effect on the reaction rates than the residue in the position N-terminal to a reactive Asx residue.[7,11] The structural features of the C-terminal residue that influence Asx reactivity include steric bulk and residue flexibility. Ionization state and the resulting catalytic and inductive/electrostatic effects of the C-terminal residue are also known to influence the degradation rates of the Asx residue.[16,61] These studies indicate that amino acid sequences in proteins where the Asx residue is succeeded by a Gly, His, Asp, or Ser residue are potential hot spots to the deamidation or isomerization pathway.[11,20,29,56] Lys residues preceding a labile Asn have been shown to increase reactivity of an Asn residue due to their potential catalytic effect on Asn deamidation rates.[62] Model peptides containing labile Asx residues formulated in lyophilized PVP solids have been studied for degradation due to Asn deamidation or Asp isomerization reactions in order to evaluate the potential effects of primary sequence on Asx reactivity in the solid state. These investigations have indicated that effects of primary sequence on Asx reactivity in the solid state are similar to the effects of primary sequence on Asx reactivity in solution.

This information on the influence of primary sequence of a protein on Asx reactivity could be used for engineering of protein molecules where the residues adjacent to a labile Asx residue could modified for the purpose of achieving stability to deamidation or isomerization reactions. The site-directed mutagenesis of residues adjacent to labile Asx may offer increased stability to a protein, especially in those where the labile Asn or the Asp residues are important for protein activity.

EFFECT OF SECONDARY STRUCTURE ON Asx REACTIVITY

Asx residues located in a region of a protein having an organized secondary structure have been shown to have significantly lower susceptibility to deamidation and isomerization reactions compared to those residues located in a random coil region of a protein.[23,63] The reactivity of an Asx residue located in an α-helical or β-turn structural region of a protein is inhibited due to a conformationally restricted environment of the protein secondary structure.[12,64,65] Lura and Schirch have demonstrated the importance of local conformation and structure on the deamidation rates of Asn-Gly sequences in studies using model tetrapeptides.[66] The inhibition of Asx reactivity due to a conformationally restricted environment has also been reported in studies comparing Asx reaction rates in linear and cyclic peptides.[67,68] Studies conducted with a model β-hairpin turn peptide have demonstrated that a significant disruption in secondary structure of the turn peptide occurs upon replacement of an Asn residue in the turn peptide with an Asp or an IsoAsp residue.[69] The deamidation of Asn^8 residue in hGRF(1–32)NH_2 led to significant disruption in the alpha helical character of the N-terminal region of the peptide. The decrease in the helicity was attributed to the breakdown of hydrogen bonding pattern in the N-terminal region of the peptide due to formation of the $IsoAsp^8$ hGRF(1–32)NH_2.[70] These results demonstrates that the deamidation of Asn residues located in helical and turn sequences of a protein is a thermodynamically unfavorable process. Mimetics of β-turn structures have been used to investigate the influence of β-turn structure and the location of the Asn residue in the β-turn on its deamidation rates.[62,68] These studies show that Asn deamidation rates of peptides are threefold faster in linear analogs than in their cyclic analogs comprising a β-turn. Within the cyclic analogs comprising a β-turn structure, the analog with the Asn residue at position 2 in the β turn deamidated 30-fold faster than the analog with the Asn at position 3 in the turn. The decreased stability of the Asn at position 2 of the turn to the deamidation reaction was attributed to the fact that the formation of cyclic imide at this particular position during the deamidation reaction resulted in an increased structural stability of the turn structure.[71] In general, the lowering of Asx reactivity in proteins where the labile Asx residue is located in a region of protein secondary structure is primarily due to the conformationally restricted environment of the region.[12,72] Formulation factors that increase the secondary structure content, that is, the α-helical or β-turn content, in a protein molecule could lead to increased stabilization of these Asx residues in proteins.[23]

EFFECT OF TERTIARY STRUCTURE ON Asx REACTIVITY

The tertiary structure that is inherent in a native protein molecule can also influence the reactivity of Asx residues.[12,73–77] A protein in the native state exhibits a tertiary structure that results from the interactions between the different secondary structural regions of the protein. The overall reduction in the conformational mobility of a peptide chain due to tertiary structure can lead to an attenuation in Asx reaction rates.[73,78] This effect of protein tertiary structure is evident when comparing Asx reactivity in native proteins to that in their truncated peptide models. An example of this phenomenon is seen in studies comparing deamidation rates of Asn residues in ribonuclease A to its peptide models. The native enzyme shows greatly reduced deamidation rates compared to its peptide models.[79] In studies investigating deamidation rates of labile Asn[67] in native and reduced form of ribonuclease A, Wearne and Creighton observed that the deamidation rates of the protein in the reduced form were 30-fold greater than the rates in the native form.[80] Similar attenuation in Asn reactivity due to protein tertiary structure has been reported for recombinant human lymphotoxin.[22] Bischoff et al.[81] have described the local conformational flexibility in a protein structure as an important determinant of Asn deamidation rate in hirudin. Studies comparing Asp reactivity in two closely related monoclonal antibodies have demonstrated that isomerization rates in these species are influenced by: (i) the conformational flexibility of the labile Asp residue and (ii) the solvent accessibility of the labile Asp residue (unpublished work). Another aspect related to the effect of tertiary structure of proteins on Asx reactivity is the presence of residues that lie within close proximity to an Asx residue due to a protein structure. These proximal residues may catalyze Asx reactivity in a protein in a manner similar to that of residues directly linked to Asx residues through a protein primary sequence.[82] In relation to Asp isomerization, Brennan et al.[30] have suggested that any protein conformation or structure that would increase the protonation of an Asp side chain would also increase its reactivity. They suggest that the increased protonation of Asp residues in proteins due to their tertiary structures could be responsible for the Asp isomerization rates in proteins being faster than the isomerization rates in their peptide models. Asp isomerization rates in certain proteins have been observed to be similar to Asn deamidation rates in spite of Asn deamidation rates being significantly faster than Asp isomerization rates in their peptide models.[40] The accelerated reactivity of the Asp residues in a protein as compared to its peptide model was noted for MAb I and its peptide model VDYDG (unpublished work). In studies investigating Asp reactivity in MAb I and VDYDG, it was noted that at pH > 6, Asp isomerization rates in the protein were in fact faster than in the peptide. A Tyr residue within H-bonding distance of the Asp residue in the protein was implicated as the reason for the observed acceleration in reaction rates for the protein.

Robinson et al.[83] have proposed a model, available at the website www.deamidation.org, for prediction of Asn deamidation rates in proteins based on the deamidation rates investigated in various peptide models. The model considers both the role of protein primary sequence and that of protein tertiary structure in influencing Asn deamidation rates. The model, which is primarily based on protein primary sequence information, incorporates a correction factor to account for the influence of protein tertiary structure on Asn reactivity.[83]

PEPTIDE MODELS TO STUDY Asx REACTIVITY

Much of the published literature addressing the issue of Asn deamidation or Asp isomerization in proteins has made use of peptide models as surrogates for understanding the reactivity of Asn and Asp residues in proteins. This approach has enabled a fundamental understanding of the mechanism of these reactions without interference from other chemical and physical degradations that are commonly observed while studying the whole protein molecule. Focusing on specific Asx residues using peptide models has enabled isolation of formulation factors that contribute to Asx reactivity. This section describes some examples of peptide models reported in the literature that were used to study Asx reactivity in proteins.

The use of the peptide model VYPNGA has contributed significantly to our understanding of Asn deamidation and formulation factors that influence Asn deamidation rates in proteins. The peptide sequence VYPNGA corresponds to residues 22 to 27 of adrenocorticotropic hormone. Studies conducted with VYPNGA have led to a better understanding of the pH dependency of Asn deamidation.[17] Using the VYPNGA peptide model, Brennan et al.[40] reported on a decrease in Asn deamidation rates in aqueous solutions containing methanol, ethanol, glycerol, etc. due to a dielectric effect of the co-solvent on Asn reactivity. The dependence of deamidation rates on solution polarity was also investigated in VYPNGA by Li et al.[41] in studies conducted in PVP- and glycerol-based aqueous solutions. Further studies involving the influence of formulation viscosity in concentrated PVP solutions (50% w/w PVP) and lyophilized solids with varying water content were also conducted by Li et al. using the peptide model VYPNGA. These investigations demonstrated that increased viscosities of formulations led to inhibition in Asn deamidation rates of peptides due to a decrease in the conformational flexibility of the reactive peptide species. The addition of poloxamer 407 and sucrose to an aqueous formulation of VYPNGA was observed to inhibit deamidation rates by as much as 40%.[45] Studies conducted with VYPNGA in buffered solutions containing sucrose and mannitol, as well as in lyophilized solids prepared from these solutions, showed a decrease in deamidation rates in the presence of these excipients both in solution and solid state.[55] A significant inhibition in deamidation rates was also reported for VYPNGA in lyophilized formulations containing PVP in both the glassy and rubbery states.[25]

Various other peptide models have been used to investigate Asx reactivity. Xie et al. used peptide models KQNSL and LSNNSL to study Asn deamidation in recombinant human lymphotoxin at positions Asn^{19} and Asn.[20,40,41] A comparison of deamidation rates between the peptide models and the protein suggested stabilization of Asn residues in the native protein due to reduced flexibility of these residues in the protein.

Goolcharran et al.[29] used the peptide GQNHH to investigate Asn deamidation at Asn^{10} residue in vascular endothelial growth factor (VEGF). Investigations using GQNHH indicated that His residues adjoining an Asn in a protein can accelerate Asn deamidation through both acid and base catalytic effects.[29] Another example of neighboring residue catalysis is seen in the peptide model CYNGQTNC for RNAse A. It has been suggested that the Lys residue preceding the Asn residue catalyzes the deamidation of the Asn residue.[62] Studies conducted with a model β-hairpin turn peptide have indicated that the formation of Asp or IsoAsp through deamidation of Asn to be thermodynamically unfavorable in β-turn structures. The use of β-turn mimetics to investigate deamidation of Asn residues located in β-turn regions of a protein demonstrates inhibition in deamidation rates due to conformational restrictions of a turn structure in a protein.[71]

The studies involving model peptides, as indicated above, show the potential advantages in using these models to aid in the understanding of deamidation and isomerization reactions in proteins. Typically, the degradation rates observed in peptide models correspond to the maximal rates possible in proteins for these degradation reactions. Studies involving peptide models thus represent accelerated conditions useful for studying Asx reactivity in a protein. The information gathered from these studies can be used to predict actual deamidation and isomerization rates in proteins.[83] The stability data obtained from the studies involving peptide models could be used to select suitable conditions for formulation of their parent proteins.

The disadvantage of using peptide models lies in the fact that predictions based on these models may not accurately describe Asx degradation rates in certain native proteins. It has been observed that the Asp isomerization rates in some proteins (e.g., MAb I) are faster than the isomerization rates predicted from their peptide models (unpublished work).[30] Tertiary structural considerations leading to possible intra-spatial neighboring effects on Asx reaction rates in proteins are not taken into account when using peptide models for predicting these rates. The deviations from the peptide model-predicted Asx reaction rates could also be due to the presence of a catalytic residue that is not directly linked to an Asx residue through the protein primary sequence. This catalytic residue may lie in close spatial proximity to the reactive Asx residue due to higher structures in the native proteins.

SUMMARY

Nonenzymatic post-translational modification of Asn and Asp residues in proteins presents a major stability challenge for successful formulation of their therapeutics. In an effort to improve the stability of the protein molecule to deamidation and/or isomerization reactions, the approach of modifying the labile Asn and Asp residues in a protein through site-directed mutagenesis is not always feasible. This approach may result in a significant compromise in potency of the protein therapeutic. In such situations, a fundamental understanding of the reaction mechanisms involved in deamidation and isomerization pathways is desired. The mechanistic understanding can be used in designing formulations that limit the instability in proteins due to Asn and Asp degradation reactions. Both Asn deamidation and Asp isomerization reactions are mechanistically similar, but the effects of formulation factors such as pH, solvent dielectric, etc. are observed to be significantly different for each reaction.

As a general rule, protein structure inhibits degradation rates of Asn and Asp due to conformational constraints on the protein backbone and the Asx side chains. The degradation rates observed using peptide models are typically representative of the maximal degradation rates attainable in a protein. The use of peptide models provides the formulator with a simple tool to understand Asn and Asp degradation in their parent protein molecules. Data gathered from studies using these peptide models provides valuable information in the choice of optimal formulation conditions for the parent proteins.

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